

The Solvability Frontier: Which Open Problems Sit Next to Solved Ones, and a Backtest of the Rule

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Abstract

We embed 5,080 problem statements across seven canon branches with a small sentence encoder, call a problem's [reach](#) its highest similarity to a solved problem, and draw a [frontier](#) at the 10th percentile of solved problems' reach, 0.781 on this set [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Open problems are sorted into a [reach class](#): 251 close to known results, 764 borderline, 911 that need a new idea and 103 in a branch with too few solved rows to judge [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. A [backtest](#) at cutoffs 2005 and 2021 scores the rule on the questions posed by those years under two outcome codings. Settled only (solved in 2026): at 2005 the inside rows settle at 21% against 6% outside (112 and 71 rows, ratio 3.804, permutation $p = 0.006$, AUC 0.800), at 2021 at 12% against 1% (238 and 126 rows, ratio 14.824, $p = 0.000$, AUC 0.883) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]; removing the solved rows that carry no resolved year, 11 and 21, drops the gaps to 13% against 6% ($p = 0.131$) and 3% against 1% ($p = 0.269$) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Settled or advanced (solved or partial): at 2005, 46% inside against 61% outside (ratio 0.752, $p = 0.051$, AUC 0.479), at 2021, 43% against 39% (ratio 1.102, $p = 0.503$, AUC 0.523) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The conclusion depends on how partial is coded, and partial is the uncertain column: the ingest assigned it in 2026 from a status page with no rule for how much progress counts. The embeddings and the 50 stored neighbours come from the 2026 corpus, so the test is a check against known outcomes under that leak; a forecast would need embeddings fitted on pre-cutoff text and a neighbour set frozen per cutoff.

1 Introduction

Bucket's solvability atlas lists open and closed problems with their statements, branch, status and, where known, the years they were posed and resolved. The question behind the frontier is whether an open problem that sits close in meaning to solved problems is closer to being solved, and whether an AI system working from known results could reach it. The frontier drawing answers the first clause by a rule; this paper tests the rule on problems whose outcome is known and publishes the ranked predictions for the open ones, with every number tagged to the file and command that reproduces it.

2 Related work

The atlas draws on the 2005 Science list of 125 questions [1, 2], the 2021 Science and Shanghai Jiao Tong University list¹, the google-deepmind formal-conjectures repository² and the Wikipedia lists of unsolved problems³. The encoder is bge-small-en-v1.5, whose model card points at the C-Pack paper [3]. Systems that generate and test scientific hypotheses end to end [4, 5] are the setting in which a reach score would be used; this paper measures only whether the score tracks outcomes.

3 Preliminaries

Definition 3.1 (Set). The set is the 71 atlas problems and the 5,009 rows of `problems-sourced.tsv`, 5,080 rows once Lean theorems are left off [empirical: `output/solvability-frontier/report/frontier.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. A top-level row is solved when its status is solved. A variant is solved only when it and its parent are both solved. Each row’s statement is embedded with bge-small-en-v1.5 at revision 5c38ec7, and its 50 most similar rows are stored with cosine similarity to three decimals [empirical: `output/solvability-frontier/report/frontier.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Definition 3.2 (Reach). The `reach` of a row is its highest similarity to a solved row other than itself, taken over its stored neighbours and its stored nearest solved row.

Definition 3.3 (Frontier). The `frontier` threshold τ is the 10th percentile of reach over the solved rows. A row with reach at or above τ is inside; below it, outside. An open row whose branch holds fewer than 10 solved rows is unsampled and counted outside.

Definition 3.4 (Growth). For a top-level row beyond the frontier, `growth` is one plus the number of beyond top-level rows among its 50 stored neighbours whose similarity to it is at or above τ : the number of rows that would move inside if it were solved. Variants neither pull nor count as pulled.

Definition 3.5 (Reach class). Let u be the 75th percentile of reach among open rows inside the frontier. An open top-level row is “close to known results” when inside with reach at or above u , “borderline” when inside below u , “needs a new idea” when beyond, and “unsampled” when its branch holds fewer than 10 solved rows. Within a class, rows rank by growth, then reach, then id; a ranking by score with unique ids has one ordering [bm:BucketMath.Ranking.rank_ordered].

Definition 3.6 (Backtest). For a cutoff year c , the solved set S_c holds every row solved under 3.1 with a resolved year at or before c . The tested set holds every row posed at or before c and outside S_c . Reach is computed as in 3.2 against S_c alone. When none of a row’s stored neighbours is in S_c , its reach is unknown and bounded above by its 50th neighbour’s similarity. The threshold τ_c is the 10th percentile of reach over the rows of S_c that have a neighbour in S_c ; the rest are counted and dropped. A tested row is inside when its reach is at or above τ_c , outside when its reach or its bound is below τ_c , and undecided otherwise; undecided rows are counted and left out of the rates. The outcome is the row’s 2026 status under two codings reported side by side: “settled” counts solved only, “settled or advanced” counts solved or partial. A solved row with no resolved year that was posed by the cutoff is tested and scored as resolved; every table repeats its first row with those rows removed. Undecided rows are also scored under two extremes, all inside and all outside, as bounds without a p -value.

¹<https://www.science.org/content/resource/125-questions-exploration-and-discovery>, read from a Wayback copy of the booklet; no DOI.

²<https://github.com/google-deepmind/formal-conjectures>, Apache-2.0; no DOI.

³https://en.wikipedia.org/wiki/Lists_of_unsolved_problems, CC BY-SA 4.0; no DOI.

4 Method

Statistics. Per cutoff and per stratum we report the inside and outside counts, the resolved rate on each side, their ratio (inside over outside, undefined when the outside rate is zero [bm:BucketMath.Marketing.ratio]) and a two-sided permutation p -value: the outcomes are shuffled 10,000 times under seed 20261006 and p is the share of shuffles whose absolute rate difference is at least the observed one, with one added to numerator and denominator [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The p -value so defined is positive and at most one (C). A stratum with fewer than 10 rows on either side reports counts only. The AUC is the Mann-Whitney probability that a resolved row’s reach exceeds an unresolved row’s, ties counting one half.

Strata. Rows are stratified by the word count of the embedded text, under 20, 20 to 40 and over 40 words, and by branch. The bands partition the naturals (C). The length bands exist because the dated lists are short paraphrases: their statements average 19 words against 34 for the rest of the file [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Reproduction. `tools/solvability-atlas/build.sh` rebuilds the embeddings and the neighbour file; `npx ts-node scripts/build-solvability-frontier-report.ts` writes `frontier.json`, `backtest.json`, `predictions.json` and `predictions.csv` under `output/solvability-frontier/report/` and the page data under `src/lib/research-os/`. `make figures` in this directory redraws every figure and table from those files.

5 Provenance and method

5.1 Sources

1 lists every source the rows came from, with its licence, what was read from it and what the ingest did to it. The 5080 rows are 71 curated atlas problems and 5009 ingested rows [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-06, `python3 tools/solvability-atlas/makeup.py -publish`]. 1 shows the make-up of the set under every label the pipeline reads: `mathematics` holds 4,112 rows, `formal-conjectures` supplies 3,567, `Apache-2.0` covers 3,557, 588 rows carry a posed year, 236 a resolved year, and 2,194 are variants of another row [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-06, `python3 tools/solvability-atlas/makeup.py -publish`].

5.2 Pipeline

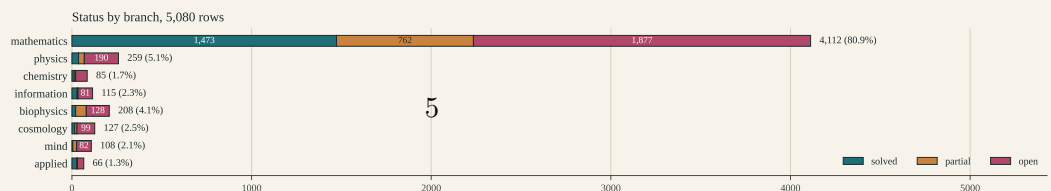
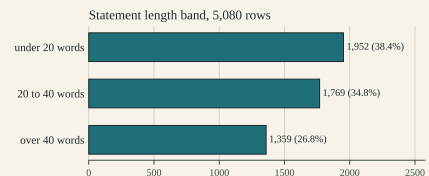
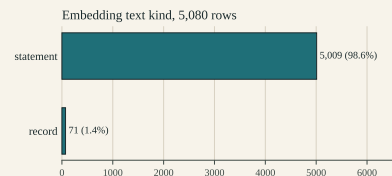
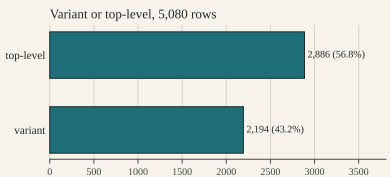
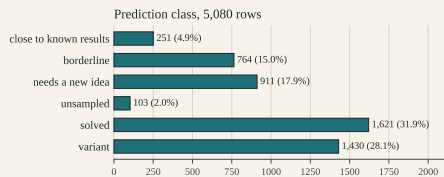
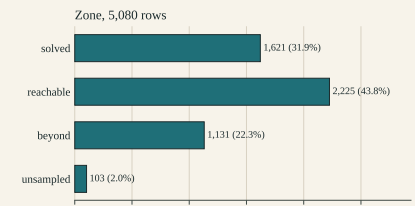
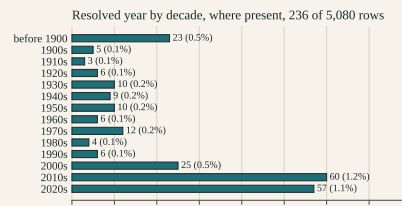
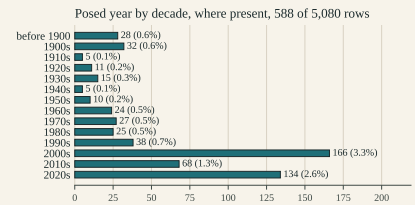
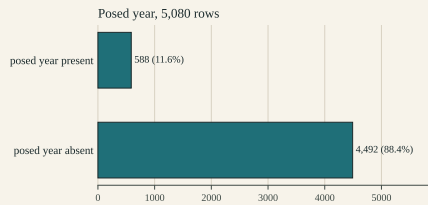
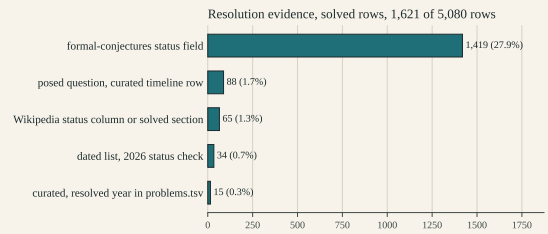
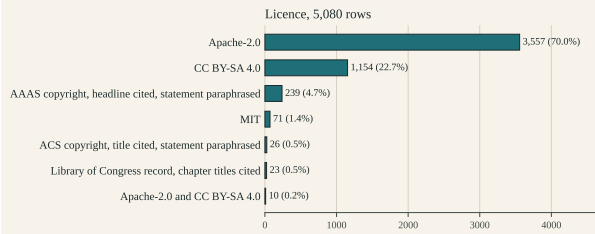
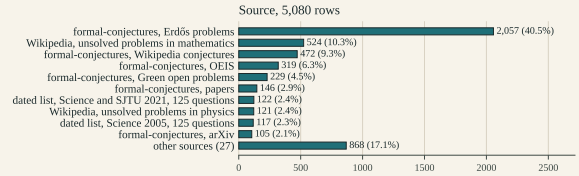
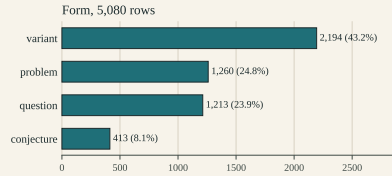
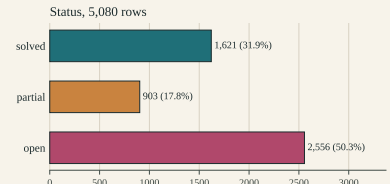
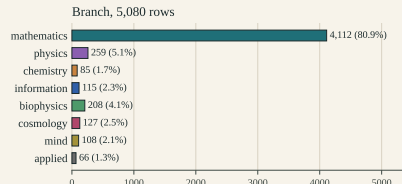
Each step names its formula; i and j index rows.

1. **Text.** t_i is the ingested statement when present, else the atlas problem record (name, aliases, statement, keywords, key work titles), else name plus keywords: 5,009 statement, 71 record and 0 name plus keywords rows [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-06, `python3 tools/solvability-atlas/makeup.py -publish`]. The branch word never enters the text.
2. **Embedding.** $e_i = f(t_i)/\|f(t_i)\|$ with $f = \text{bge-small-en-v1.5}$ at revision 5c38ec7, 384 dimensions, cached by model revision and text hash [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Table 1: Sources of the rows, with licence, what was taken and what was done [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-06, python3 tools/solvability-atlas/makeup.py – publish].

Source	Rows	Licence	What was taken	What was done to it
formal-conjectures	3567	Apache-2.0 (3557); Apache-2.0 and CC BY-SA 4.0 (10)	statement (text field, or the Lean declaration when empty), status, family and declaration suffix as name; 84 posed and 39 resolved years	deduped by normalised title; suffixed declarations marked as variants of their file’s top-level row; keywords mined from the statement; years from the text only
Wikipedia unsolved lists	992	CC BY-SA 4.0	full list item as statement, bold or linked head as name, section heading as status; 18 posed and 54 resolved years	branch per page, market per list, deduped against the atlas and within the file; years from the text only
Hilbert’s problems	23	CC BY-SA 4.0	explanation cell, table status and year columns, 1900 as posed year; 23 posed and 6 resolved years	partial for partial, no consensus, disputed, weaker form or partially cells; curator overrides on 14 and 18
Smale’s problems	14	CC BY-SA 4.0	explanation cell, table status and year columns, 1998 as posed year; 14 posed and 0 resolved years	same table rule; curator overrides on 8, 14 and 17
dated lists (Science 2005 and 2021, DARPA, XPRIZE, neuroscience 23, Holy Grails)	325	AAAS copyright, headline cited, statement paraphrased (239); CC BY-SA 4.0 (37); ACS copyright, title cited, statement paraphrased (26); Library of Congress record, chapter titles cited (23)	headline under 15 words as name, list year as posed year, a 2026 status check per row; 325 posed and 34 resolved years	statement paraphrased in 15 to 40 words; deduped by title and at 0.85 similarity; solved rows carry the discovery year
solved timelines, Wikipedia and Kavli	88	CC BY-SA 4.0	a question shown to predate its answer, the discovery or proof year, a posed-evidence quote under 12 words; 53 posed and 88 resolved years	statement rewritten as the question stood before resolution; discovery rows with no prior question held back; 159 pages verified
atlas, problems.tsv	71	MIT	name, branch, level, Lean status, posed and resolved years, markets, keywords, all curated; 71 posed and 15 resolved years	resolved year marks solved; embedding text is the problem record when one exists

Make-up of the 5,080 rows behind the solvability frontier, built 2026-10-06



3. **Similarity.** $s_{ij} = e_i \cdot e_j$, the cosine. The 50 highest s_{ij} per row are stored to three decimals, with the nearest solved row over the whole set [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].
4. **Solved set.** $S = \{i : \text{status}_i = \text{solved} \wedge (\text{parent}_i = \emptyset \vee \text{status}_{\text{parent}_i} = \text{solved})\}$, so a proved special case of an open problem stays partial; $|S| = 1,621$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].
5. **Reach.** $r_i = \max_{j \in S, j \neq i} s_{ij}$ over the stored neighbours and the stored nearest solved row (3.2).
6. **Threshold.** $\tau = Q_{0.1}(\{r_j : j \in S\})$, the value at index $\lfloor 0.1|S| \rfloor$ of the sorted solved reaches; $\tau = 0.781$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].
7. **Zones.** solved if $i \in S$; unsampled if the branch holds fewer than 10 solved rows; reachable if $r_i \geq \tau$; beyond if $r_i < \tau$: 1,621, 103, 2,225 and 1,131 rows [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].
8. **Radius.** With $\text{span}(x, a, b) = \min(1, \max(0, (x-a)/(b-a)))$: $\rho_i = 0.6(1 - \text{span}(r_i, \min_S r, 1))$ for solved rows, $0.6 + 0.4(1 - \text{span}(r_i, \tau, 1))$ for reachable rows, $1 + 0.5(1 - \text{span}(r_i, \text{floor}, \tau))$ for beyond rows and $1 + 0.5(1 - \text{span}(r_i, \text{floor}, 1))$ for unsampled rows, where $\text{floor} = \min(\tau, \min_i r_i)$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].
9. **Growth.** $g_i = 1 + |\{j \in N_i : \text{zone}_j = \text{beyond}, j \text{ top-level}, s_{ij} \geq \tau\}|$ over the 50 stored neighbours N_i (3.4).
10. **Angle.** $p_i = (e_i - \bar{e}) V_2$ with V_2 the first two right singular vectors of the centred embeddings; θ_i is the rank of $\text{atan2}(p_{i2}, p_{i1})$ over the rows, scaled to $[0, 2\pi)$ [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].
11. **Reach class.** $u = Q_{0.75}(\{r_i : i \text{ open, inside}\}) = 0.854$ [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]; close when $r_i \geq u$, borderline when $\tau \leq r_i < u$, needs a new idea when $r_i < \tau$, unsampled by branch (3.5).
12. **Backtest statistic.** At cutoff c : $d_c = p_{\text{in}} - p_{\text{out}}$ with p the resolved share of scored rows on each side of τ_c ; ratio $p_{\text{in}}/p_{\text{out}}$, undefined when $p_{\text{out}} = 0$ [bm:BucketMath.Marketing.ratio]; $\text{AUC} = \Pr(r_{\text{resolved}} > r_{\text{unresolved}})$ with ties at one half (3.6).
13. **Permutation test.** $p = (1 + |\{b : |d_b^*| \geq |d_c|\}|)/(B + 1)$ with $B = 10,000$ shuffles of the outcomes across the inside and outside rows under seed 20261006 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts], so $0 < p \leq 1$ (C).

OpenAlex activity. A title-and-abstract search on each sourced row’s name or first clause, run against the OpenAlex works index with a key, stores the matching work count, counts per year for the last 30 years and the three most cited works per row in `tools/solvability-atlas/openalex-sourced.json`. The free daily budget is 1,000 requests, two per row. The first run took the file in its own order and stored 499 of the 5,009 rows before the budget ran out, 388 of them with at least one matching work, all 499 in mathematics (median 15 works) and none in the other seven branches [empirical: output/solvability-frontier/report/makeup/makeup.json, 2026-10-06, python3 tools/solvability-atlas/makeup.py].

–publish]; a second run the same day was refused for ten hours. Later runs take the remaining rows in an order that interleaves the branches. Until every branch holds at least 30 rows with data no comparison across branches is possible. The frontier score does not use these counts yet; 1 carries no activity axis and the distribution by branch sits in the page’s make-up charts.

6 Results

6.1 The frontier

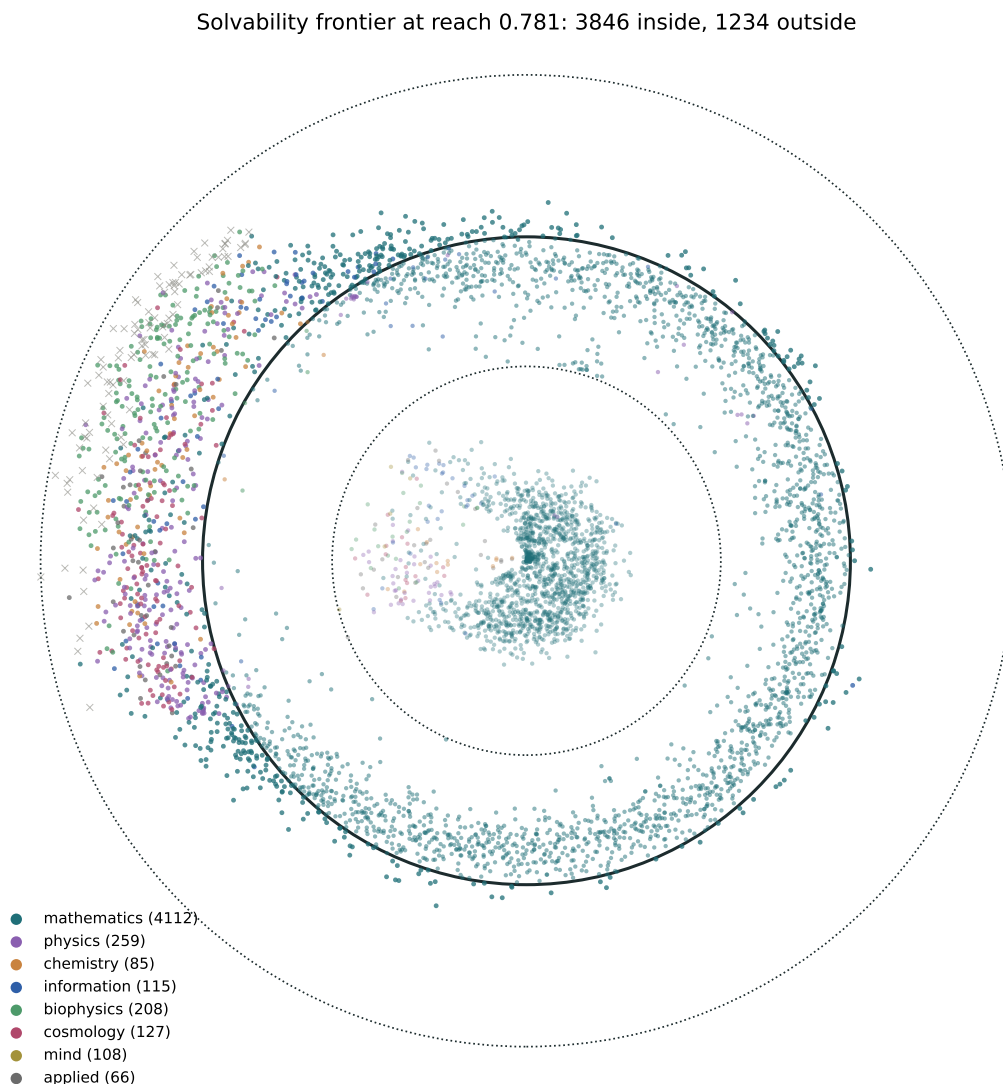


Figure 2: The frontier on 5,080 rows. Solved rows fill the inner disc by reach, open rows inside fill the ring up to the circle at 0.781, outside rows sit beyond it by how far their reach falls short, and unsampled rows are grey crosses [empirical: `output/solvability-frontier/report/frontier.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. Angle is the row’s rank along the first two principal components of the embeddings.

The frontier sits at $\tau = 0.781$: 1,621 rows solved, 2,225 reachable, 1,131 beyond and 103 unsampled, 3,846 inside and 1,234 outside [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. 2 gives the split per branch. Mathematics holds 4,112 of the 5,080 rows and 1,473 of the 1,621 solved ones [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts], so it sets the threshold on its own.

Table 2: Rows per branch and zone [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Branch	Total	Solved	Reachable	Beyond	Unsampled	Inside	Outside
mathematics	4112	1473	2171	468	0	3644	468
physics	259	37	28	194	0	65	194
biophysics	208	21	4	183	0	25	183
cosmology	127	16	0	111	0	16	111
information	115	29	19	67	0	48	67
mind	108	5	0	0	103	5	103
chemistry	85	13	3	69	0	16	69
applied	66	27	0	39	0	27	39

6.2 Backtest

3, 4, 5, 6 and 3 hold the backtest. At cutoff 2005, 108 rows were solved by then, 16 of them with no solved neighbour among the 50 stored and dropped from the threshold, which came out at 0.644; 287 rows were posed by 2005 and open at the cutoff, 38 unsampled, 66 undecided and 183 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. At cutoff 2021, 190 rows were solved by then, 11 dropped, threshold 0.656; 465 rows tested, 58 unsampled, 43 undecided and 364 scored [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Settled only. At 2005 the inside rows settle at 21% over 112 against 6% over 71 outside, ratio 3.804, $p = 0.006$, AUC 0.800 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. At 2021 they settle at 12% over 238 against 1% over 126, ratio 14.824, $p = 0.000$, AUC 0.883 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The settled rows number 28 at 2005 and 29 at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Of the tested rows, 32 at 2005 and 33 at 2021 are solved today with no resolved year, 11 and 21 of them scored; with those removed the inside rate falls to 13% against 6% (ratio 2.285, $p = 0.131$, AUC 0.700) and to 3% against 1% (ratio 4.065, $p = 0.269$, AUC 0.700) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Most of the settled-only gap is those rows.

Settled or advanced. At 2005 the inside rate is 46% against 61% outside, ratio 0.752, $p = 0.051$, AUC 0.479; at 2021 it is 43% against 39%, ratio 1.102, $p = 0.503$, AUC 0.523 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. With the undated solved rows removed: 40% against 61% ($p = 0.008$) and 37% against 39% ($p = 0.818$) [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Undecided rows. The 66 undecided rows at 2005 resolve at 32% settled and 53% settled or advanced; the 43 at 2021 at 28% and 53% [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Placing all of them inside or all outside gives, settled only, inside rates of 25% or 21% against 6% or 18% at 2005 and 14% or 12% against 1% or 8% at 2021; settled or advanced, 48% or 46% against 61% or 57% at 2005 and 45% or 43% against 39% or 43% at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The last rows of each table carry the exact counts.

Table 3: Backtest at cutoff 2005, settled only, by stratum; the last two rows place the undecided rows all inside and all outside [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	112	71	21%	6%	3.804	0.006
all, undated solved rows removed	101	71	13%	6%	2.285	0.131
under 20 words	28	20	7%	0%		0.503
20 to 40 words	44	36	14%	6%	2.455	0.282
over 40 words	40	15	40%	13%	3.000	0.102
applied	4	3	under 10 on one side			
biophysics	18	44	11%	4%	2.444	0.577
chemistry	7	2	under 10 on one side			
cosmology	11	5	under 10 on one side			
mathematics	49	1	under 10 on one side			
physics	23	16	30%	12%	2.435	0.263
all undecided inside	178	71	25%	6%	4.487	
all undecided outside	112	137	21%	18%	1.174	

Table 4: Backtest at cutoff 2005, settled or advanced, by stratum [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	112	71	46%	61%	0.752	0.051
all, undated solved rows removed	101	71	40%	61%	0.654	0.007
under 20 words	28	20	46%	70%	0.663	0.139
20 to 40 words	44	36	46%	75%	0.606	0.013
over 40 words	40	15	45%	13%	3.375	0.053
applied	4	3	under 10 on one side			
biophysics	18	44	78%	73%	1.069	0.756
chemistry	7	2	under 10 on one side			
cosmology	11	5	under 10 on one side			
mathematics	49	1	under 10 on one side			
physics	23	16	39%	31%	1.252	0.745
all undecided inside	178	71	48%	61%	0.798	
all undecided outside	112	137	46%	57%	0.800	

Under settled or advanced, two strata reach p below 0.05 and run against each other: at 2005 the 20 to 40 word band resolves at 46% inside against 75% outside ($p = 0.013$), while at 2021 the over 40 word band resolves at 45% inside against 0% outside over 20 rows ($p = 0.000$) [empirical:

Table 5: Backtest at cutoff 2021, settled only, by stratum [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	238	126	12%	1%	14.824	0.000
all, undated solved rows removed	217	126	3%	1%	4.065	0.269
under 20 words	101	57	1%	0%		1.000
20 to 40 words	77	49	8%	2%	3.818	0.253
over 40 words	60	20	35%	0%		0.002
applied	13	11	0%	0%		1.000
biophysics	39	52	3%	0%		0.431
chemistry	30	16	3%	0%		1.000
cosmology	20	10	0%	0%		1.000
information	18	11	6%	9%	0.611	1.000
mathematics	87	1	under 10 on one side			
physics	31	25				1.000
all undecided inside	281	126	14%	1%	17.936	
all undecided outside	238	169	12%	8%	1.529	

Table 6: Backtest at cutoff 2021, settled or advanced, by stratum [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Stratum	Inside	Outside	Resolved in	Resolved out	Ratio	p
all	238	126	43%	39%	1.102	0.503
all, undated solved rows removed	217	126	37%	39%	0.960	0.818
under 20 words	101	57	37%	37%	0.994	1.000
20 to 40 words	77	49	49%	57%	0.864	0.465
over 40 words	60	20	45%	0%		0.000
applied	13	11	15%	18%	0.846	1.000
biophysics	39	52	64%	67%	0.952	0.823
chemistry	30	16	20%	19%	1.067	1.000
cosmology	20	10	40%	40%	1.000	1.000
information	18	11	33%	9%	3.667	0.197
mathematics	87	1	under 10 on one side			
physics	31	25				0.223
all undecided inside	281	126	44%	39%	1.144	
all undecided outside	238	169	43%	43%	1.006	

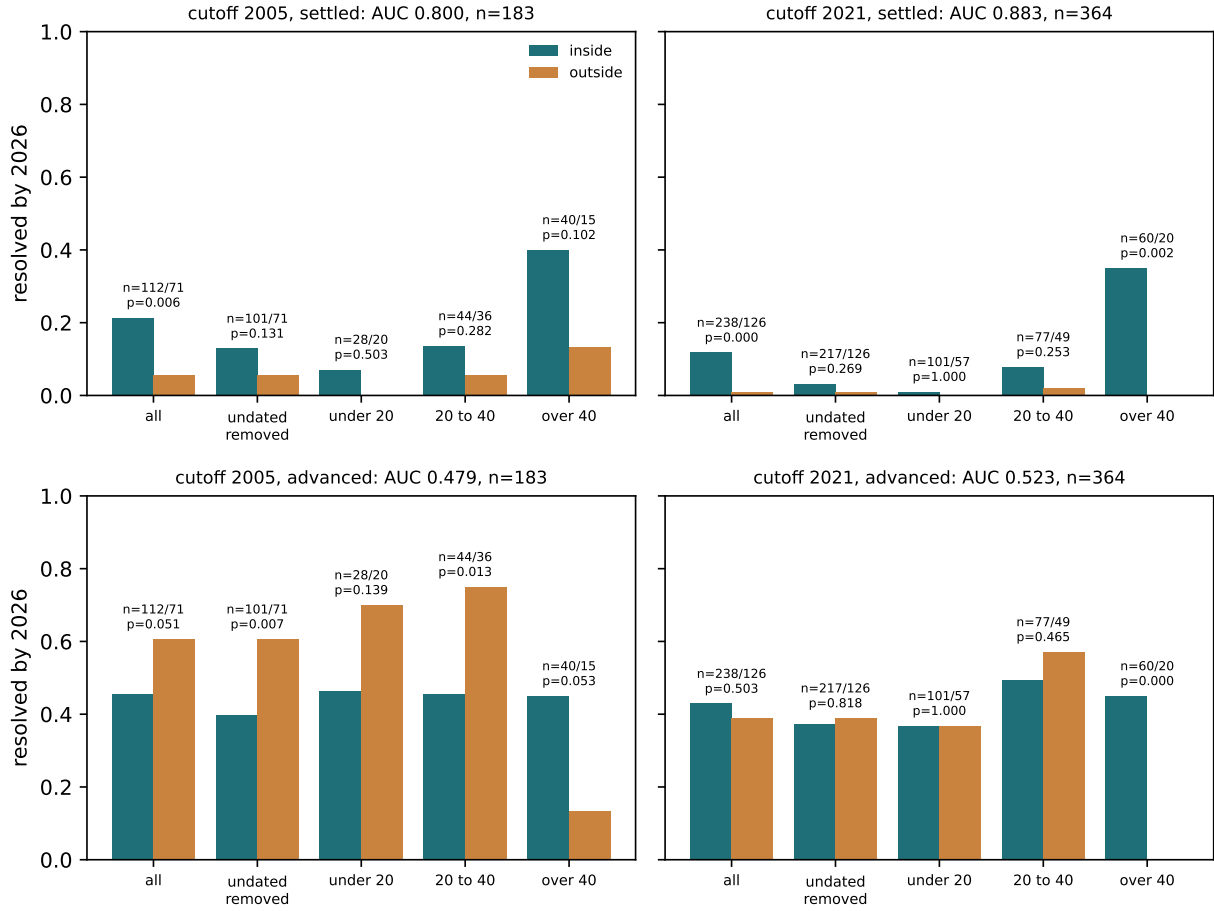


Figure 3: Resolved rate by 2026 inside and outside the cutoff frontier under both codings, over all scored rows, with the undated solved rows removed, and by length band [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The second is a cohort effect: the long statements outside at 2021 are the 2021 list’s own longer paraphrases, all still open five years on [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Mathematics scores one row outside at both cutoffs [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts] because nearly every dated mathematics row has a solved mathematics neighbour, so no mathematics rate is given.

6.3 Where this is weak

Partial. The two codings disagree on sign at 2005 and on size at 2021. Partial was assigned in 2026 by the ingest from a status page, with no threshold for how much progress counts, so the column that decides the result is the least reliable one in the file.

Leakage. The embeddings and the 50 stored neighbours come from the 2026 corpus. Statements written after a cutoff, including the 2021 list when scoring 2005, shape the geometry each cutoff is scored on, and the nearest solved row at a cutoff was found among neighbours chosen in 2026. The test checks the rule against known outcomes under that leak, which is a weaker claim than a forecast. A clean version needs embeddings fitted on text dated at or before the cutoff and a neighbour set frozen per cutoff.

Vocabulary. Cosine similarity between statement embeddings measures shared words and phrasing. Two problems can share a method and no vocabulary, or vocabulary and no method.

Mathematics. 4,112 of 5,080 rows and 1,473 of 1,621 solved rows are mathematics [empirical: output/solvability-frontier/report/frontier.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. The threshold is a mathematics quantity, and the other six branches are read against it.

Two cohorts. Of the 517 rows with a posed year, 241 come from the 2005 and 2021 Science lists, and those two lists supply most of the tested rows at both cutoffs [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. A list’s paraphrases share length and style, which is what the length bands are for, and the over 40 word result at 2021 shows the cohort rather than the rule [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

The 50-neighbour bound. At a cutoff with few solved rows, most of a row’s solved peers fall outside its 50 stored neighbours, so its reach is unknown: 66 rows undecided and 16 solved rows dropped at 2005, 43 and 11 at 2021 [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts]. Dropping them can only raise the threshold; the undecided bounds above show how far they can move the inside rate.

Resolved years. Only 224 of the 1,621 solved rows carry a resolved year, so the solved set at a cutoff, 108 at 2005 and 190 at 2021, is a fraction of what was solved by then [empirical: output/solvability-frontier/report/backtest.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts], and 32 and 33 solved rows with no year sit in the tested set instead.

6.4 Predictions

7 gives the class counts: 251 rows close to known results, 764 borderline, 911 that need a new idea and 103 unsampled, with $u = 0.854$ [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. 8, 9, 10 and 11 list the first ten of each class by growth; the full ranked table with the three nearest solved problems and the starting works is `predictions.csv`. B lists the 56 open atlas problems in full. Among them, Goldbach and the Langlands program are close to known results, with reach 0.867 and 0.863 to an Erdős problem and to the geometric Langlands conjecture [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]; the Riemann hypothesis, Navier-Stokes, dark matter, weak cosmic censorship and quantum gravity head the “needs a new idea” class by growth, 4, 4, 3, 3 and 3 [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The Goldbach placement is the vocabulary effect of 6.3: the nearest solved row shares the words, and nothing in the method says whether it shares a method.

Table 7: Open top-level problems per reach class [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

Class	Open problems
close to known results	251
borderline	764
needs a new idea	911
unsampled	103

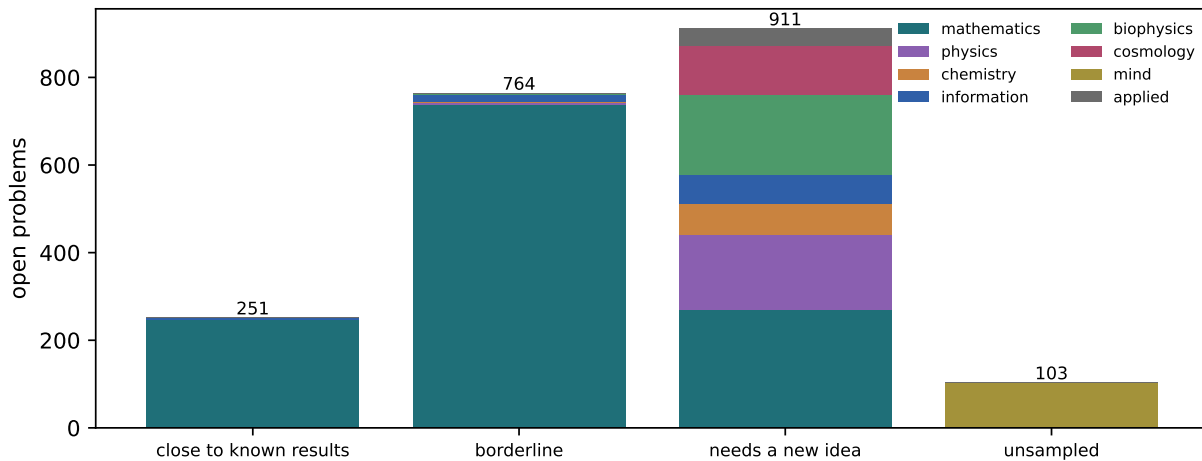


Figure 4: Reach classes by branch [empirical: `output/solvability-frontier/report/predictions.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`].

7 Calibration

The central claims are empirical, so the calibration is the backtest itself: the rule is fixed before the outcomes are read, the outcomes are the 2026 status column with its per-row source, and

Table 8: Close to known results, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
Pierpont Prime, infinitely many pierpont primes	mathematics	0.870	4	Regular Primes, infinitude of irregularprimes
Can the graph isomorphism problem be solved in polynomial time on a classical computer?	information	0.924	3	Graph isomorphism in quasi-polynomial time
Wieferich Prime, infinite isWieferichPrime	mathematics	0.871	3	Regular Primes, infinitude of irregularprimes
OEIS A945, every prime occurs	mathematics	0.868	2	OEIS A363102, a eq one or prime
Factorial Prime, infinitely many factorial primes	mathematics	0.908	1	Regular Primes, infinitude of irregularprimes
Caratheodory conjecture	mathematics	0.894	1	Caratheodory conjecture for surfaces of smoothness $C^{\{4\}}$ (Brendan Guil)
Is graph canonization polynomial-time equivalent to the graph isomorphism problem?	information	0.887	1	Graph isomorphism in quasi-polynomial time
Erdos problem 1030	mathematics	0.882	1	Erdos problem 1014
Erdos problem 968	mathematics	0.882	1	Erdos problem 822
OEIS A93818	mathematics	0.878	1	OEIS A372761, exists unique a eq prime

Table 9: Borderline, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
The density hypothesis for zeroes of the Riemann zeta function	mathematics	0.795	6	Erdos problem 12, erdos sarkozy density 0
Grand Riemann hypothesis	mathematics	0.789	6	Poisson Conjecture
Are there infinitely many Pell primes?	mathematics	0.804	5	Regular Primes, infinitude of irregularprimes
Can integer factorization be done in polynomial time?	mathematics	0.804	5	Linear programming in polynomial time
Are there infinitely many Wolstenholme primes?	mathematics	0.796	5	Regular Primes, infinitude of irregularprimes
Are there infinitely many Kynea primes?	mathematics	0.782	5	Regular Primes, infinitude of irregularprimes
The covering problem of Rado	mathematics	0.824	4	Erdos problem 1071, i
Hilbert?Polya conjecture	mathematics	0.820	4	OEIS A1818, conjecture1
Woodin's Ω -conjecture	mathematics	0.811	4	Erdos problem 1036, alon hajnal
Are there infinitely many Wieferich primes?	mathematics	0.809	4	OEIS A159829, conjecture1

Table 10: Needs a new idea, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
Cosmological principle	physics	0.721	8	Baryon acoustic oscillations
Extended Riemann Hypothesis	mathematics	0.778	6	Riemann Zeta Values, infinite irrational at odd
Generalized Riemann hypothesis for Selberg class	mathematics	0.777	6	Erdos problem 193, z2
L = P problem	information	0.758	6	Erdos problem 318, univ
Dark energy	physics	0.663	6	Photon underproduction crisis (2014?2015)
Riemann Hypothesis, riemannHypothesis	mathematics	0.776	5	OEIS A49473
Generalized Riemann hypothesis	mathematics	0.773	5	Erdos problem 990, hayman
Are there infinitely many sexy primes?	mathematics	0.769	5	Regular Primes, infinitude of irregularprimes
Supermassive black holes	physics	0.757	5	Black hole at the Galactic center
Black holes, black hole information paradox, and black hole radiation	physics	0.740	5	Ultraviolet catastrophe

Table 11: Unsampled, first ten by growth [empirical: output/solvability-frontier/report/predictions.json, 2026-10-06, npx ts-node scripts/build-solvability-frontier-report.ts].

Problem	Branch	Reach	Growth	Nearest solved
What is reflected in sensory neocortical activity?	mind	0.724	0	Neurons that encode location
Where do our memories get stored and how are they retrieved again?	mind	0.713	0	Cellular basis of memory storage
Is there a canonical computation performed by cortical columns?	mind	0.701	0	Erdos problem 298, natural density
Can human memory be stored, manipulated, and transplanted digitally?	mind	0.700	0	Cellular basis of memory storage
Connectionism	mind	0.699	0	Keller's conjecture (Joshua Brakensiek, Marijn Heule, John Mackey, Dav
Modularity of mind	mind	0.699	0	Cellular basis of memory storage
How Are Memories Stored and Retrieved?	mind	0.697	0	Cellular basis of memory storage
Free will, particularly the neuroscience of free will	mind	0.696	0	Keller's conjecture (Joshua Brakensiek, Marijn Heule, John Mackey, Dav
Can we understand the action of brain in natural environments?	mind	0.695	0	Chemical neurotransmission at synapses
How do neurons compute?	mind	0.692	0	Chemical neurotransmission at synapses

the threshold and the strata are recomputed at each cutoff from rows dated at or before it. The error budget is the permutation p -value, the stratum floor, the two codings and the two bounded exclusions. Under settled only the inside rows settle at a higher rate at both cutoffs, and most of that gap is the solved rows with no resolved year; under settled or advanced the AUC sits at 0.479 and 0.523, each within 0.03 of chance [empirical: `output/solvability-frontier/report/backtest.json`, 2026-10-06, `npx ts-node scripts/build-solvability-frontier-report.ts`]. The reading is that the result turns on the partial column, and no forecast claim survives the embedding leak of 6.3.

8 Limitations

6.3 lists the weak points of the backtest and the frontier: the partial column, the 2026 embedding leak, vocabulary as similarity, mathematics dominance, the two cohorts, the 50-neighbour bound and the missing resolved years.

Sources and licences. `problems.tsv` is MIT. formal-conjectures rows are Apache-2.0. Wikipedia lists and the DARPA and XPRIZE rows are CC BY-SA 4.0 with attribution in the data. The Science 2005 and 2021 lists, the systems-neuroscience chapter titles and the Holy Grails titles are cited as headlines under 15 words with paraphrased statements, since the bodies are under copyright. The encoder is MIT.

References

- [1] Donald Kennedy and Colin Norman. “What Don’t We Know?” In: *Science* 309.5731 (2005), p. 75. DOI: [10.1126/science.309.5731.75](https://doi.org/10.1126/science.309.5731.75).
- [2] Science News Staff. “So Much More to Know”. In: *Science* 309.5731 (2005), pp. 78–102. DOI: [10.1126/science.309.5731.78b](https://doi.org/10.1126/science.309.5731.78b).
- [3] Shitao Xiao et al. *C-Pack: Packed Resources For General Chinese Embeddings*. The bge-small-en-v1.5 model card points at this paper. 2023. arXiv: [2309.07597](https://arxiv.org/abs/2309.07597) [cs.CL].
- [4] Juraj Gottweis et al. *Accelerating scientific discovery with an AI co-scientist*. 2025. arXiv: [2502.18864](https://arxiv.org/abs/2502.18864).
- [5] Chris Lu et al. “The AI Scientist: towards fully automated open-ended scientific discovery”. In: (2024). arXiv: [2408.06292](https://arxiv.org/abs/2408.06292).

A Glossary

reach <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts>

A problem’s highest cosine similarity to a solved problem other than itself, over its stored neighbours; see 3.2.

frontier <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts>

The reach value at the 10th percentile of solved problems’ reach; a problem at or above it is inside, below it outside; see 3.3.

growth <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-frontier.ts>

For a problem beyond the frontier, one plus the number of beyond problems among its stored neighbours that sit at or above the threshold from it; see 3.4.

reach class <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-predictions.ts>

One of four labels for an open problem: close to known results, borderline, needs a new idea, unsampled; see 3.5.

backtest <https://github.com/gianyrox/bucket-foundation/blob/dev/src/lib/research-os/solvability-backtest.ts>

The frontier rule applied at a past cutoff year to the problems posed by then, scored against their 2026 status; see 3.6.

B The atlas problems

Problem	Class	Branch	Reach	Growth	Nearest solved
Goldbach conjecture	reach	mathematics	0.867	0	Erdos problem 418, conditional
Langlands program	reach	mathematics	0.863	0	Geometric Langlands conjecture
Graph isomorphism complexity	borderline	information	0.833	0	Graph isomorphism in quasi-polynomial time
Jacobian conjecture	borderline	mathematics	0.807	0	Poisson Conjecture, jacobian implication
Collatz conjecture	borderline	mathematics	0.786	0	OEIS A306424, forty three is max
Riemann hypothesis	needs a new idea	mathematics	0.771	4	Deterministic polynomial-time primality testing
Navier-Stokes existence and smoothness	needs a new idea	physics	0.671	4	Poincare, smooth known cases
Dark matter identity	needs a new idea	cosmology	0.678	3	Baryon acoustic oscillations
Weak cosmic censorship	needs a new idea	cosmology	0.659	3	Baryon acoustic oscillations
Quantum gravity	needs a new idea	cosmology	0.615	3	Microscopic Weighting, iff finite concentration
Room-temperature superconductivity mechanism	needs a new idea	physics	0.603	3	Element predicted as eka-aluminium
Programmed versus stochastic aging	needs a new idea	biophysics	0.602	3	Carrier of hereditary information
Glass transition	needs a new idea	chemistry	0.601	3	Continuum hypothesis independence
Integer factoring complexity	needs a new idea	information	0.752	2	OEIS A382590, kthPrimeFactor periodic
Hadwiger-Nelson problem	needs a new idea	mathematics	0.743	2	Erdos problem 780
Invariant subspace problem	needs a new idea	mathematics	0.727	2	Green open problem 49
Matrix multiplication exponent omega	needs a new idea	information	0.725	2	OEIS A108306
Existence of one-way functions	needs a new idea	information	0.711	2	Erdos problem 198, concrete
Hubble tension	needs a new idea	cosmology	0.707	2	Expansion of the universe
BQP vs BPP separation	needs a new idea	information	0.693	2	Graph isomorphism in quasi-polynomial time
Yang-Mills mass gap	needs a new idea	physics	0.693	2	Microscopic Weighting, iff finite concentration

Problem	Class	Branch	Reach	Growth	Nearest solved
Black hole information paradox	needs a new idea	cosmology	0.691	2	Ultraviolet catastrophe
Origin of life	needs a new idea	biophysics	0.688	2	Urea synthesis from inorganic reagents
Cosmological constant problem	needs a new idea	cosmology	0.668	2	Erdos problem 1005, constant
Quantum measurement problem	needs a new idea	physics	0.637	2	Einstein problem (David Smith, Joseph Samuel Myers, Craig S
Birch and Swinnerton-Dyer conjecture	needs a new idea	mathematics	0.764	1	Elliptic Curve Rank, goldfeld two le
abc conjecture	needs a new idea	mathematics	0.760	1	Erdos problem 494, k eq 3 card gt 6
Unique Games Conjecture	needs a new idea	information	0.751	1	Complexity of computing Nash equilibria
Twin prime conjecture	needs a new idea	mathematics	0.745	1	OEIS A91669, prime and primitive root of dvd
P vs NP	needs a new idea	information	0.732	1	Complexity of computing Nash equilibria
Protein folding pathway kinetics	needs a new idea	biophysics	0.721	1	Protein structure prediction
Hodge conjecture	needs a new idea	mathematics	0.714	1	Poincare conjecture
Hadamard matrix conjecture	needs a new idea	mathematics	0.706	1	Gerstenhaber Three Matrices, exists finrank adjoin quadruple
Succinct proofs without trusted setup optimality	needs a new idea	information	0.702	1	Erdos problem 967, tauberian
Elliptic curve discrete log hardness	needs a new idea	information	0.697	1	Modularity Conjecture
Exact exchange-correlation functional	needs a new idea	chemistry	0.683	1	Erdos problem 298, natural density
Bioelectric pattern control	needs a new idea	biophysics	0.681	1	Chemical neurotransmission at synapses
Market efficiency as computational hardness	needs a new idea	information	0.674	1	Erdos problem 298, natural density
Fermion sign problem	needs a new idea	physics	0.666	1	Monochromatic Quantum Graph, eqSystem6 no solution d5
Anomalous properties of water	needs a new idea	chemistry	0.659	1	Proof of the atomic theory
Climate sensitivity bound	needs a new idea	physics	0.653	1	Sensitivity conjecture
Autoformalization of mathematics	needs a new idea	information	0.649	1	21 149, kourouka 21 149 not inner
Optimal multi-item auction design	needs a new idea	information	0.649	1	Complexity of computing Nash equilibria
2D Hubbard model ground state	needs a new idea	physics	0.640	1	Existence of time crystals (2012?2016)
Formal reasoning limits of transformers	needs a new idea	information	0.637	1	Erdos problem 152
Mitochondrial redox and aging mechanism	needs a new idea	biophysics	0.634	1	Carrier of hereditary information
AI alignment verification	needs a new idea	information	0.624	1	Pandemic Response XPRIZE
Rough volatility pricing	needs a new idea	mathematics	0.622	1	Pricing of options
Turbulence closure	needs a new idea	physics	0.617	1	Voronovskaja Type Formula, bezier bernstein operators
Rational catalyst design	needs a new idea	chemistry	0.598	1	Complexity of computing Nash equilibria
Plasma confinement stability	needs a new idea	physics	0.588	1	Solar abundance problem (1998?2022)
Light, circadian biology and metabolism	needs a new idea	biophysics	0.572	1	OEIS A100478
Smale's 18th problem limits of intelligence	unsampled	mind	0.673	0	Hilbert's tenth problem
Neural binding problem	unsampled	mind	0.660	0	Controlling neurons with light
Free energy principle formalization	unsampled	mind	0.659	0	Erdos problem 298, natural density

Problem	Class	Branch	Reach	Growth	Nearest solved
Hard problem of consciousness	unsampled	mind	0.641	0	Keller's conjecture (Joshua Brakensiek, Marijn Heule, John M

C Lean listings

`lean/Bucket/Backtest.lean`, checked by `make lean`, holds the two facts the method leans on: the corrected permutation p has a positive numerator no larger than its denominator, and the length bands cover every word count.

```

structure PermutationP where
  hits : Nat
  permutations : Nat
  bounded : hits <= permutations

def PermutationP.numerator (p : PermutationP) : Nat := p.hits + 1
def PermutationP.denominator (p : PermutationP) : Nat := p.permutations
+ 1

theorem PermutationP.numerator_pos (p : PermutationP) : 0 < p.numerator
:=
  Nat.succ_pos p.hits

theorem PermutationP.numerator_le_denominator (p : PermutationP) :
  p.numerator <= p.denominator :=
  Nat.succ_le_succ p.bounded

def lengthBand (words : Nat) : LengthBand :=
  if words < 20 then .short else if words <= 40 then .middle else .long

theorem lengthBand_total (words : Nat) :
  lengthBand words = .short \ / lengthBand words = .middle \ /
  lengthBand words = .long

```

lake build output, quoted verbatim [empirical: papers/solvability-frontier/lean, 2026-10-06, make lean].

Build completed successfully (4 jobs).